SDA Vision / Research report

earth-image.jpg



Assessment summary

Field	Recorded detail
Finding	Leaning authentic
Recorded verdict	authentic likely
Confidence	medium
Indicative rating	4/100
Rating basis	median of available model ratings; not a calibrated probability
Explanation	3 models lean authentic with matching verdicts and ratings. Review the other findings, original source and context alongside this result.
Coverage	image - 1 frame(s) analysed
Review context	Research assessment. Review alongside source information and context. When credentials are missing, origin remains unresolved by this check. Use these findings to inform a documented human review.

Provider overview

Provider	Type / status	Assessment	Rating
Claude Vision	vision / ok	authentic likely	4/100
OpenAI Vision	vision / ok	authentic likely	2/100
Gemini Vision	vision / ok	authentic likely	25/100
C2PA Content Credentials	provenance / ok	inconclusive	n/a
Local forensic cues	forensic / ok	partially synthetic	47/100

Provider evidence / 1

Field	Recorded detail
Provider	Claude Vision
Model	claude-opus-5-5
Status	ok
Assessment	authentic likely
Confidence	high
Role in assessment	Supporting
Provider summary	synthetic likelihood 4/100; tells: visually consistent with the widely documented Apollo 17 'Blue Marble' photograph (NASA AS17-148-22727, Dec 1972), geographically accurate coastlines: Africa, Arabian Peninsula, Red Sea, Gulf of Aden, Madagascar, Antarctic ice cap, physically coherent cloud systems and cyclonic swirls with plausible meteorological structure, film-scan grain and slight color cast consistent with analog photography
Evidence 1	Visually consistent with the widely documented Apollo 17 'Blue Marble' photograph (NASA AS17-148-22727, Dec 1972)
Evidence 2	Geographically accurate coastlines: Africa, Arabian Peninsula, Red Sea, Gulf of Aden, Madagascar, Antarctic ice cap
Evidence 3	Physically coherent cloud systems and cyclonic swirls with plausible meteorological structure
Evidence 4	Film-scan grain and slight color cast consistent with analog photography
Evidence 5	Natural limb shading and terminator-free full-disc illumination consistent with sun behind the camera
Evidence 6	No repeated textures, smeared detail, or generative blending artifacts at land/cloud boundaries
Evidence 7	Caveat: Visual match to a famous archival image is not proof of an unaltered copy; subtle retouching, recoloring, or a high-fidelity regenerated replica cannot be ruled out without comparison to the NASA source scan and metadata/provenance checks.

Provider evidence / 2

Field	Recorded detail
Provider	OpenAI Vision
Model	gpt-6-astra
Status	ok
Assessment	authentic likely
Confidence	high
Role in assessment	Supporting
Provider summary	synthetic likelihood 2/100; tells: Coherent African, Arabian Peninsula, and Madagascar coastlines, Intricate, nonrepeating cloud bands and storm spirals, Consistent atmospheric haze and foreshortening toward Earth's limb, Fine grain and slight softness consistent with a scanned photograph
Evidence 1	Coherent African, Arabian Peninsula, and Madagascar coastlines
Evidence 2	Intricate, nonrepeating cloud bands and storm spirals
Evidence 3	Consistent atmospheric haze and foreshortening toward Earth's limb
Evidence 4	Fine grain and slight softness consistent with a scanned photograph
Evidence 5	Composition closely resembles the historical Blue Marble photograph
Evidence 6	Caveat: Visual resemblance cannot establish provenance; compare with the archival original to check for subsequent edits.

Provider evidence / 3

Field	Recorded detail
Provider	Gemini Vision
Model	gemini-3.8-flash
Status	ok
Assessment	authentic likely
Confidence	medium
Role in assessment	Supporting
Provider summary	synthetic likelihood 25/100; tells: authentic film grain, consistent atmospheric limb scatter, historically documented cloud formations, natural optical sensor characteristics
Evidence 1	Authentic film grain
Evidence 2	Consistent atmospheric limb scatter
Evidence 3	Historically documented cloud formations
Evidence 4	Natural optical sensor characteristics
Evidence 5	Caveat: High-profile historical images can be digitally remastered or simulated by generative models.

Content Credentials / provenance

Field	Recorded detail
Provider	C2PA Content Credentials
Model	Not recorded
Status	ok
Assessment	inconclusive
Confidence	Not recorded
Role in assessment	Additional finding
Provider summary	Embedded credentials: none found. Origin remains unresolved by this check.
Evidence 1	Coverage: credentials embedded in the submitted file. External credentials were not retrieved.
Evidence 2	Verifier: c2pa-python 0.37.12; native SDK 0.91.0; specification target 2.4; policy embedded-offline-reviewed-trust-v3.
Evidence 3	Offline embedded-manifest verification; no network trust-list, manifest or revocation refresh.
Evidence 4	No reviewed local C2PA trust snapshot installed. Provenance overrides are disabled.

Provider evidence / 5

Field	Recorded detail
Provider	Local forensic cues
Model	Not recorded
Status	ok
Assessment	partially synthetic
Confidence	low
Role in assessment	Additional finding
Provider summary	Heuristic synthetic likelihood 47/100 from metadata + pixel statistics.
Evidence 1	No camera EXIF (common in AI exports, screenshots, and re-saves).
Evidence 2	ELA mean 0.5, peak 10 (uniform high ELA can indicate generation/heavy edit).
Evidence 3	High-frequency noise std 4.61 (very low/very uniform noise is a synthesis cue).
Evidence 4	Spectral peakiness 25.63 (periodic frequency peaks can mark generated textures).

Evidence supporting this verdict

Field	Recorded detail
Claude Vision	synthetic likelihood 4/100; tells: visually consistent with the widely documented Apollo 17 'Blue Marble' photograph (NASA AS17-148-22727, Dec 1972), geographically accurate coastlines: Africa, Arabian Peninsula, Red Sea, Gulf of Aden, Madagascar, Antarctic ice cap, physically coherent cloud systems and cyclonic swirls with plausible meteorological structure, film-scan grain and slight color cast consistent with analog photography
OpenAI Vision	synthetic likelihood 2/100; tells: Coherent African, Arabian Peninsula, and Madagascar coastlines, Intricate, nonrepeating cloud bands and storm spirals, Consistent atmospheric haze and foreshortening toward Earth's limb, Fine grain and slight softness consistent with a scanned photograph
Gemini Vision	synthetic likelihood 25/100; tells: authentic film grain, consistent atmospheric limb scatter, historically documented cloud formations, natural optical sensor characteristics

Additional findings

Field	Recorded detail
Local forensic cues	Supporting observation; considered alongside the model assessments.
C2PA Content Credentials	Embedded credentials: none found. Origin remains unresolved by this check.

How this assessment was reached

Field	Recorded detail
1. override	outcome=none; detail=No confirmed C2PA AI assertion or SynthID watermark. Origin remains unresolved by these checks.
2. model_agreement	synthetic=[]; authentic=['Claude Vision', 'OpenAI Vision', 'Gemini Vision']; uncertain=[]; forensic_supports_synthetic=False
3. verdict	value=authentic_likely; confidence=medium; reason=3 models lean authentic with matching verdicts and ratings. Review the other findings, original source and context alongside this result.

File and analysis record

Field	Recorded detail
File	earth-image.jpg
Analysis ID	afa06964-dc8c-4c81-8fc7-b6179d9b2705
Generated	2026-09-26T22:31:03+00:00
Tool	SDA Vision 0.1.0
Models	claude-opus-5-5, gemini-3.8-flash, gpt-6-astra
Method version	pdf-preparation-2026-09-26.1
Analysis lens	general
Original SHA-256	a4bad2aba040fd992aff7483eee6166943e232061b82d7149b2060786719753b
Ingestion note 1	Fresh provider replies acquired for eleven-examples-refresh-20260926-r1; report assembled locally from the saved responses without further API calls.
Ingestion note 2	Provider latency values are restored from the acquisition receipts (maximum sampled-frame latency per provider); raw replies and assessment results are unchanged by this bookkeeping step.